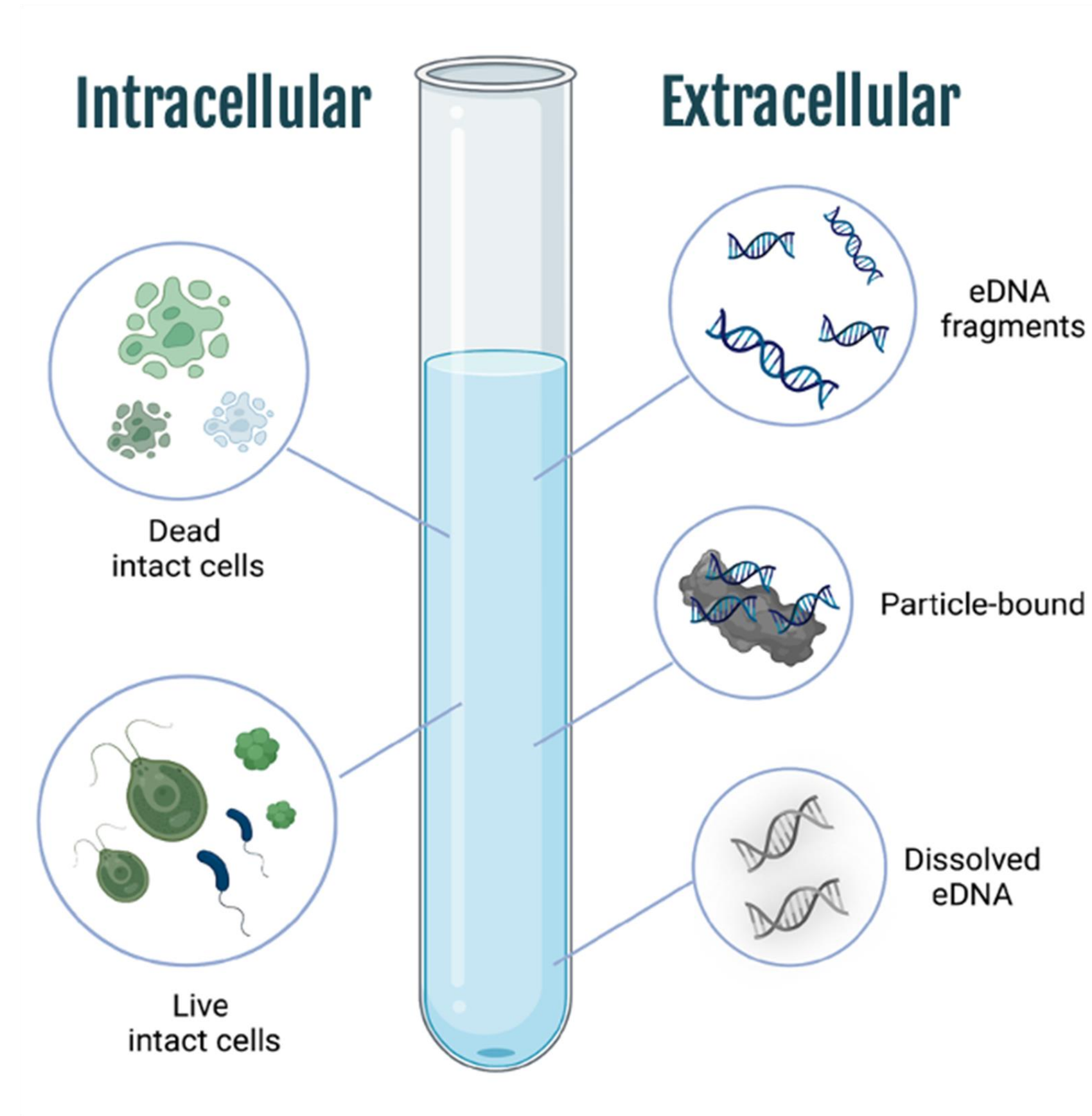
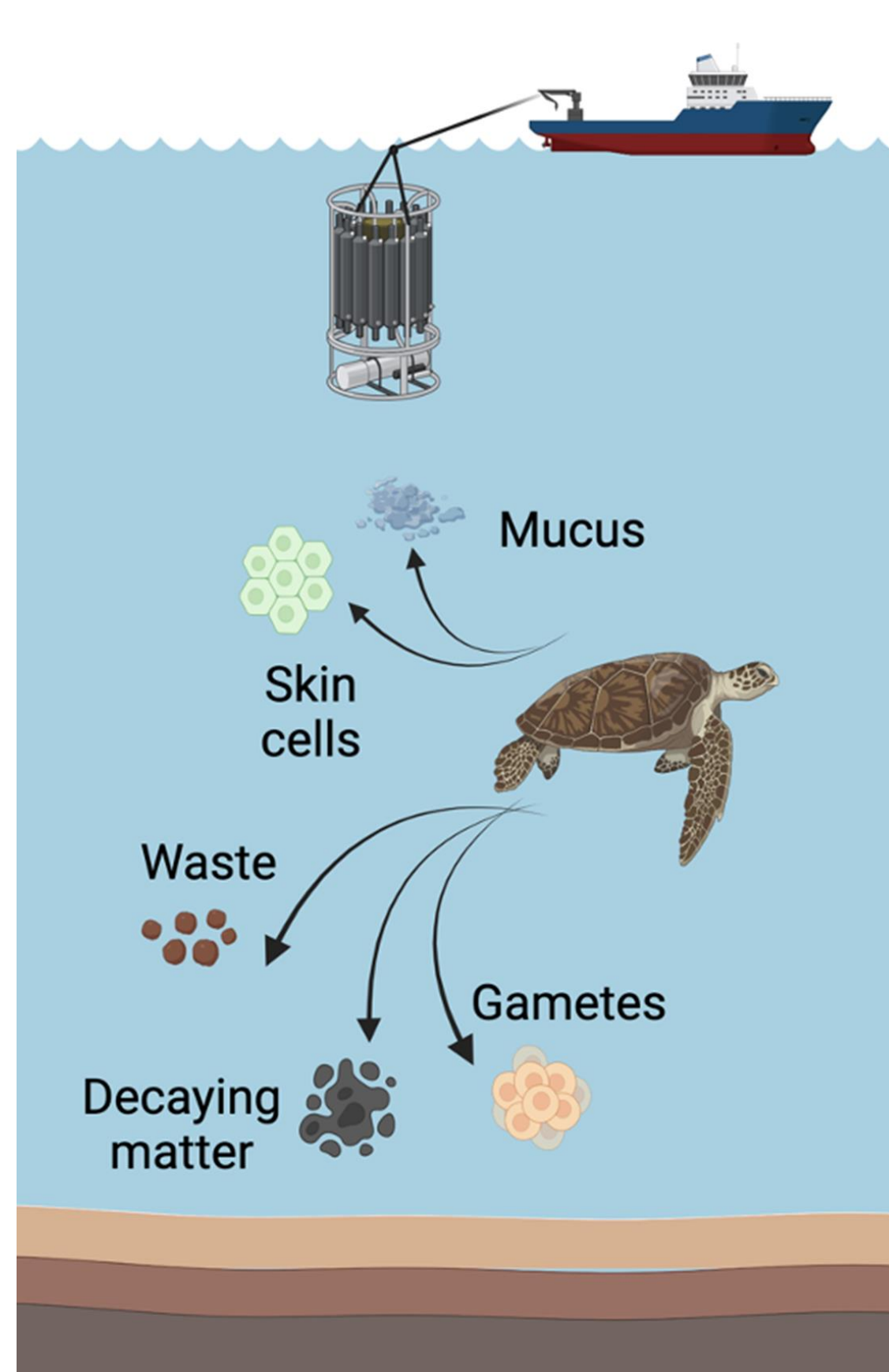


Biodiversity observation ain't easy

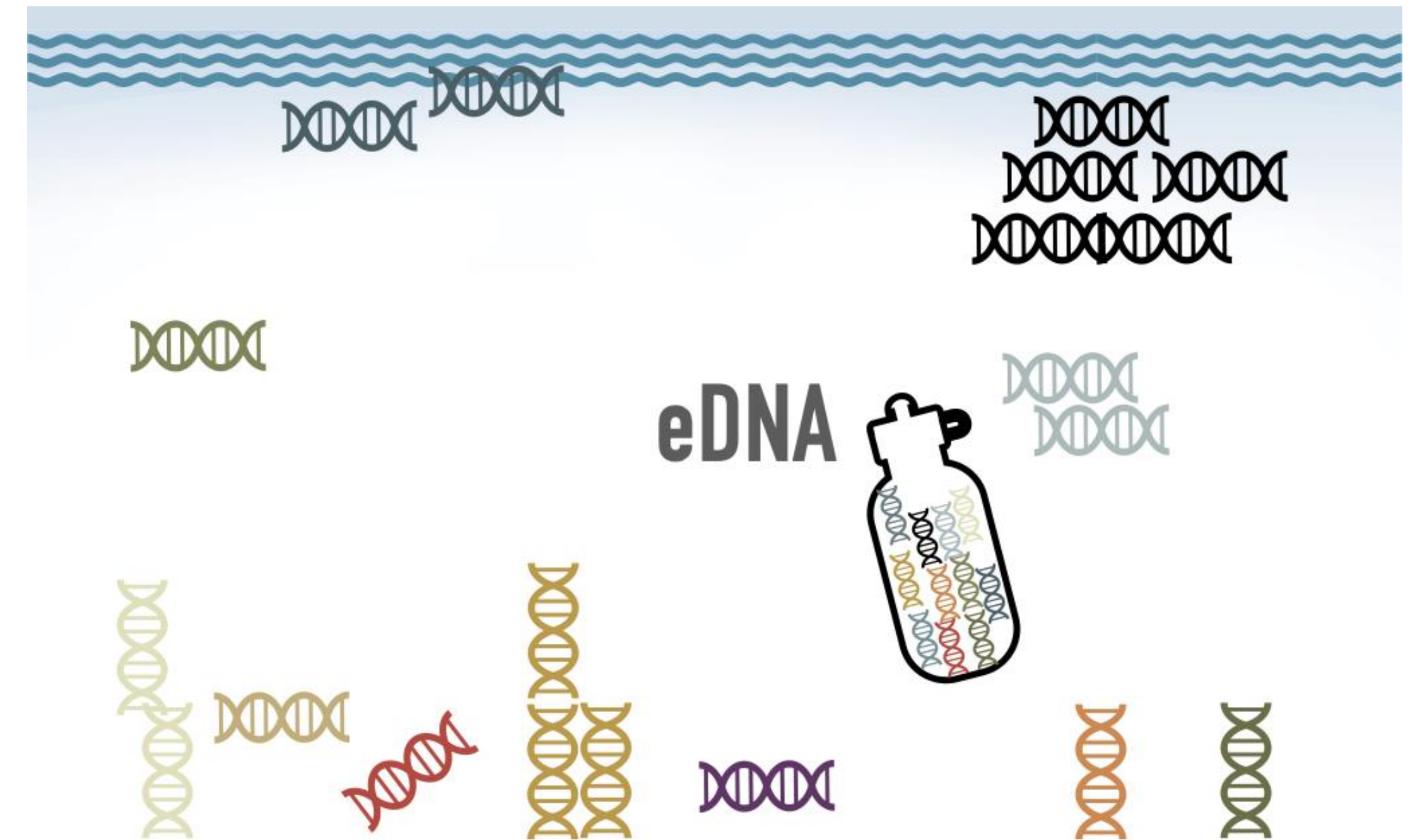
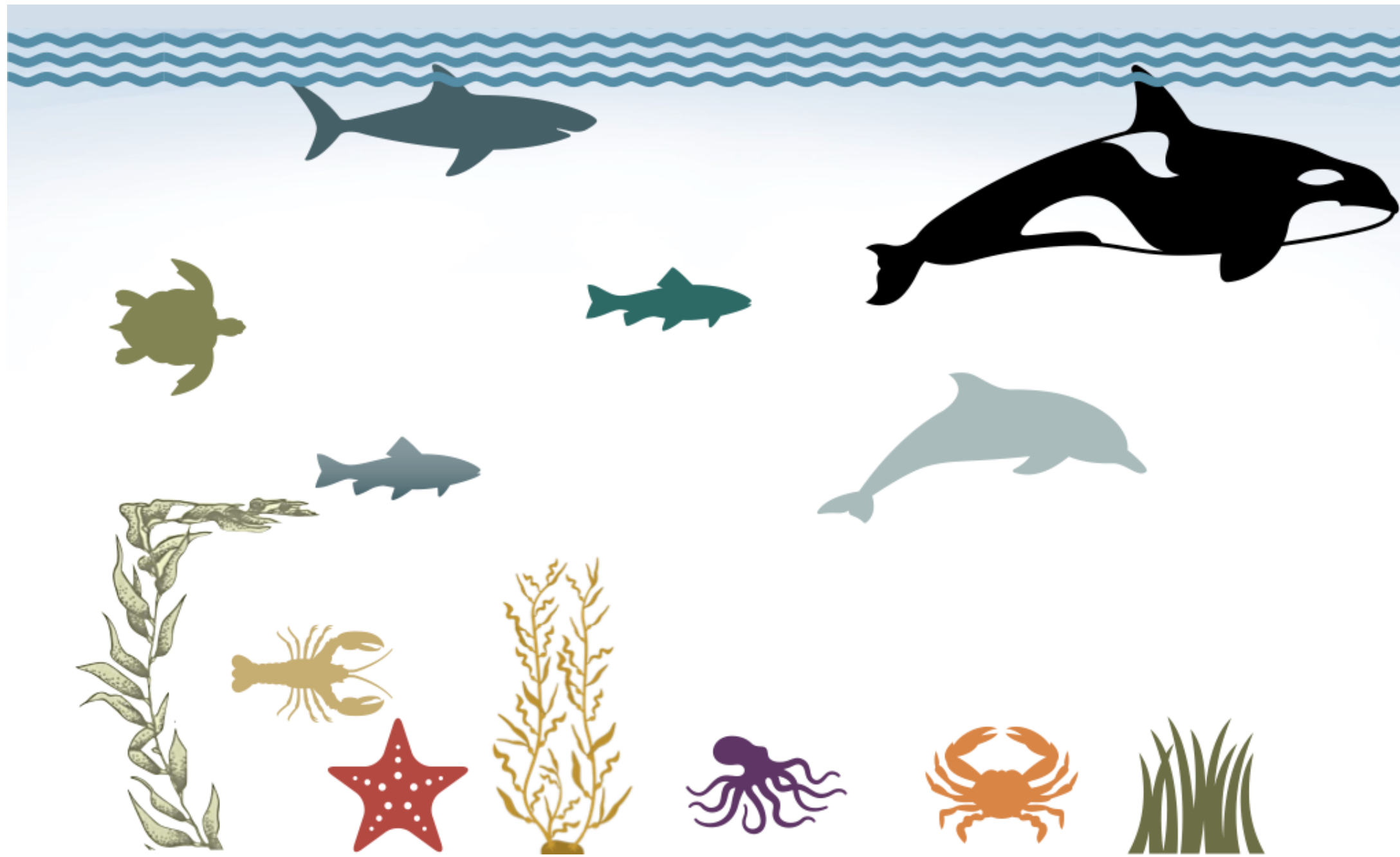
- Time and Labor intensive
- Expensive
- Requires taxonomic expertise
- Dominated by conspicuous or commercial species
- Conditions dependent

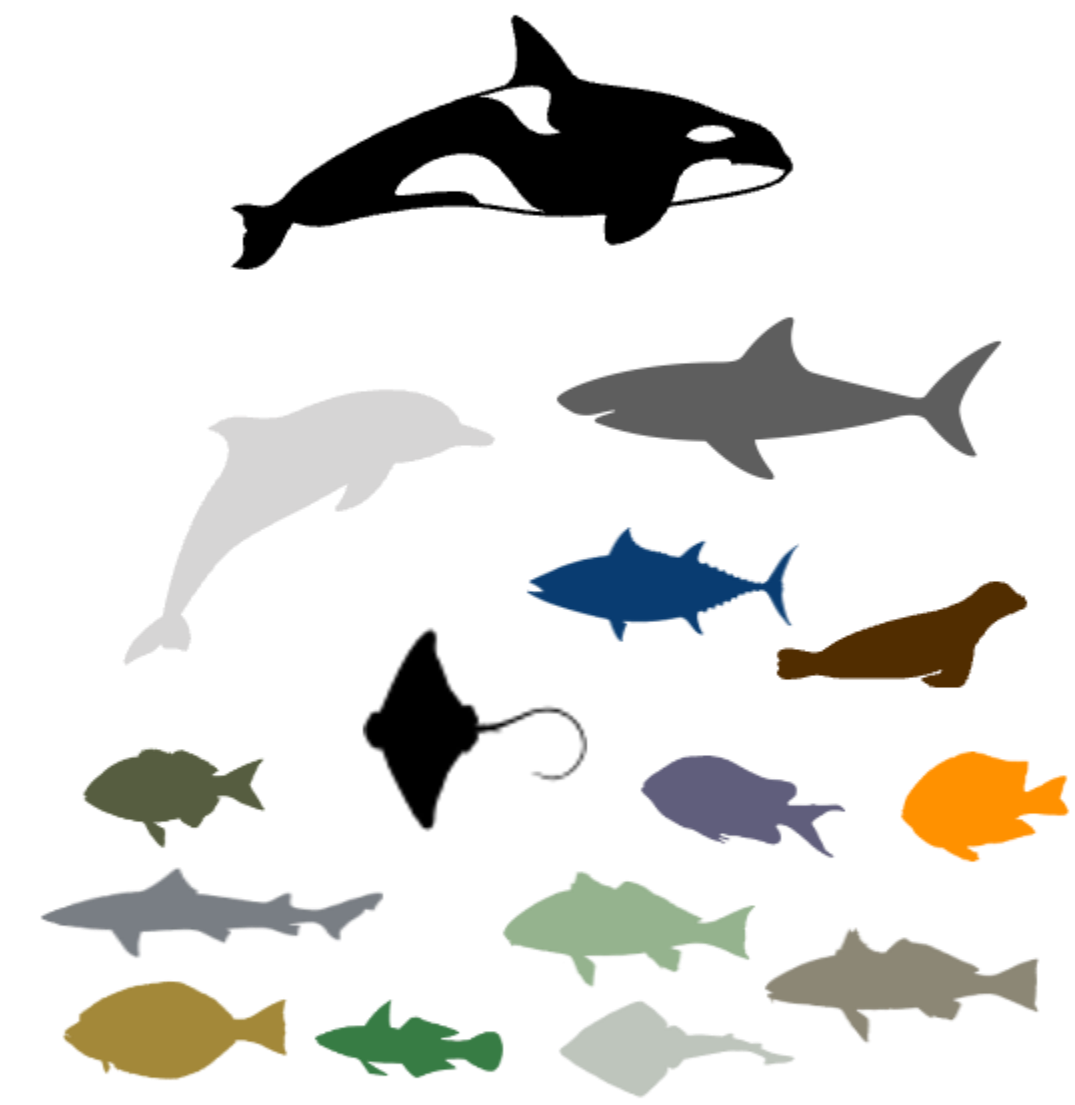
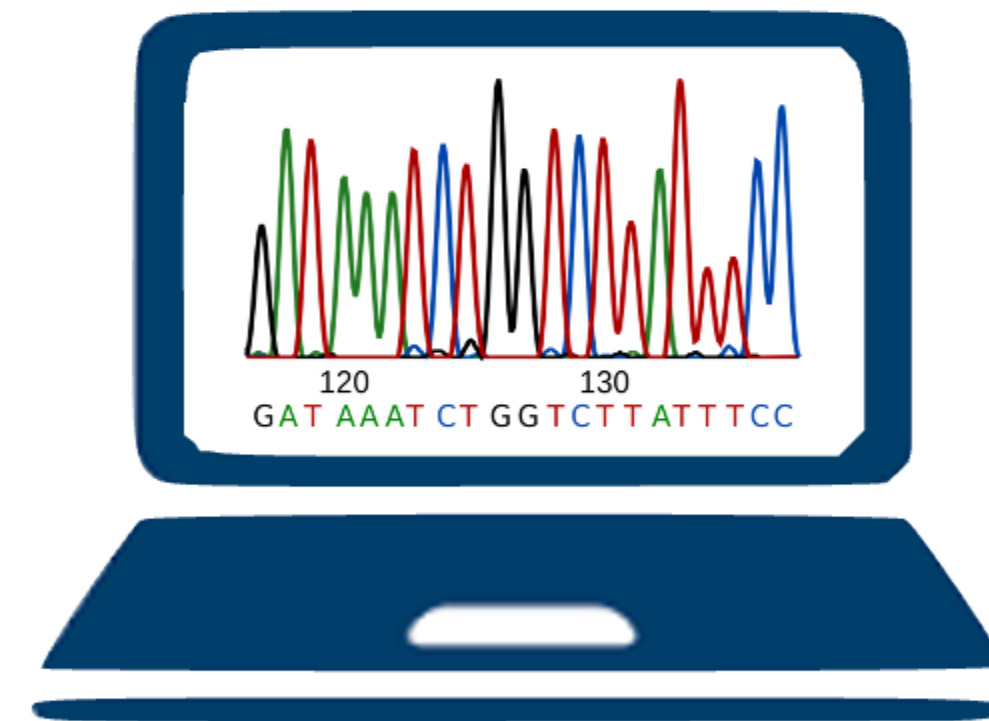
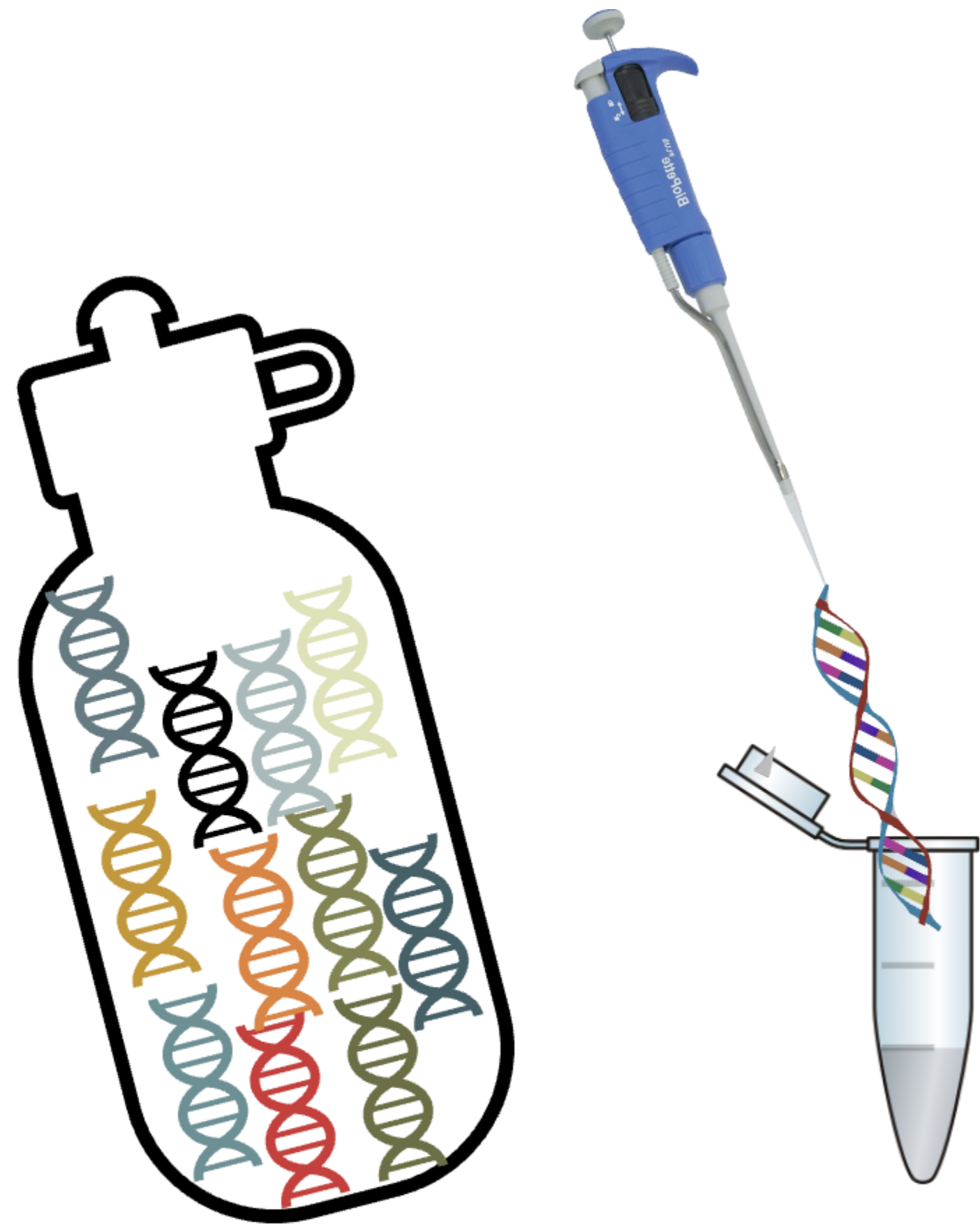


There are many types of evidence of biodiversity.



eDNA can reduce the resources needed to observe biodiversity





Sample → **Process** → **Sequence** → **Reconstruct**



Strengths of eDNA



Sampling

Ease of sampling, collect samples remotely

Minimal environmental impact

Filters, DNA extracts, and sequences can be preserved long term, allowing reanalysis



High Sensitivity

Improved detection of rare and cryptic species

Detect broad diversity of species simultaneously, from microbes to mammals



Standardization of biases

Ability to standardize taxonomic analyses

Minimize biases and increase comparability of data across sampling programs



Burden

Low cost per sample

Fast data turnaround times



Declining Cost of Sequencing

2001

\$10,000,000.00

Per DNA Gigabase

Declining Cost of Sequencing

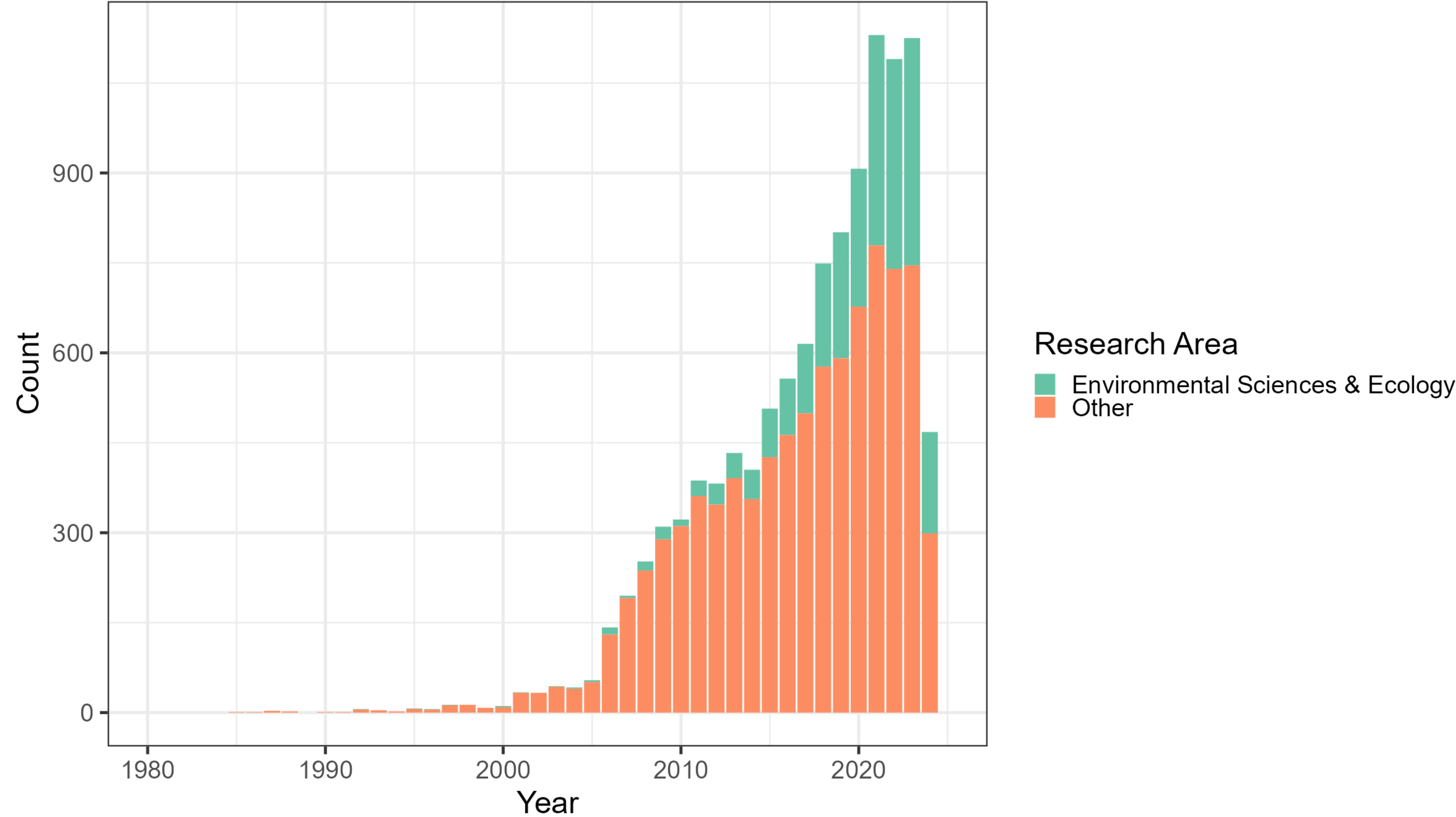
2022

\$2.00

Per DNA Gigabase

Abundance of eDNA Publications Over Time

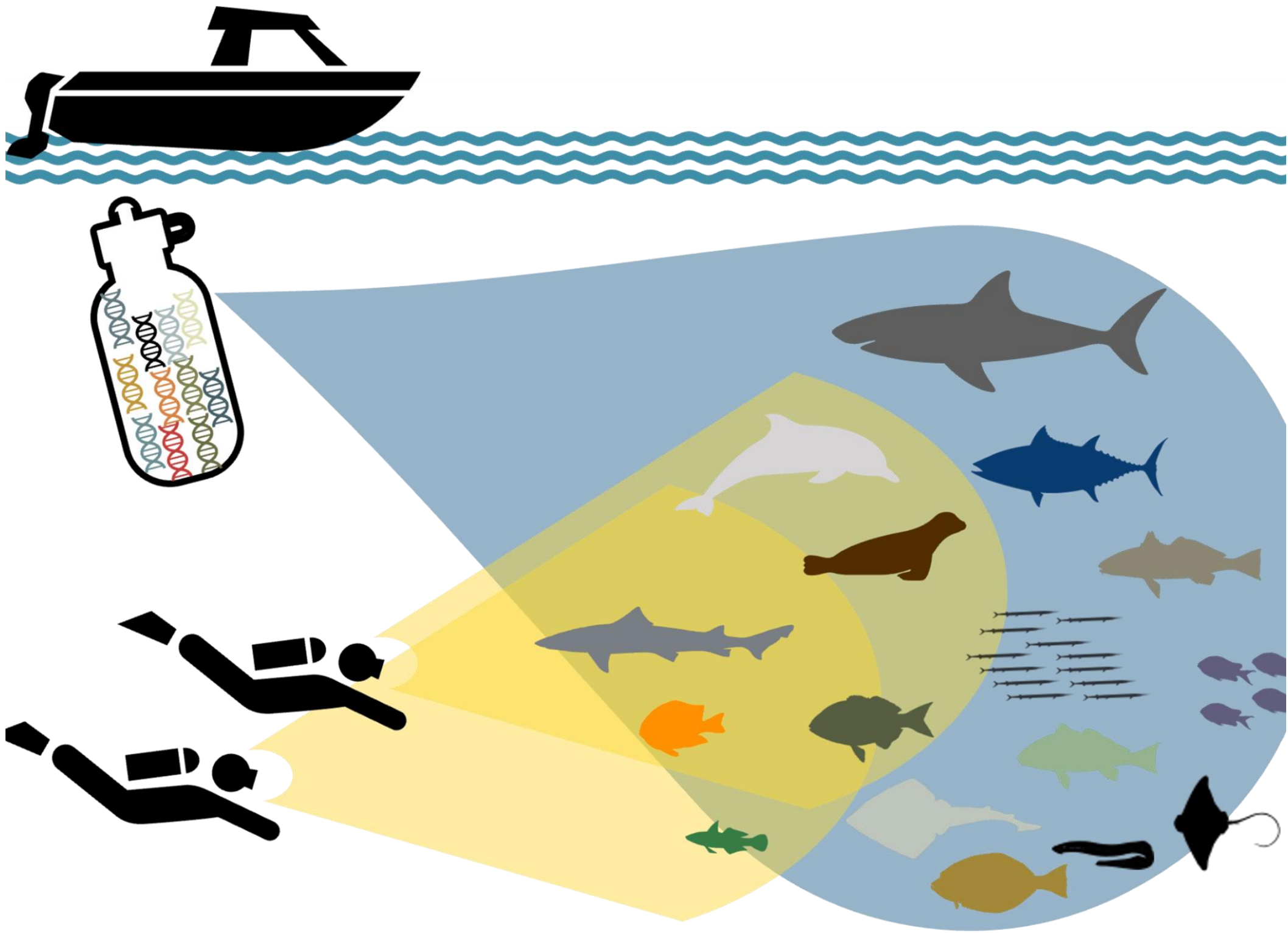
based on Chavez et al. 2021, Figure 3



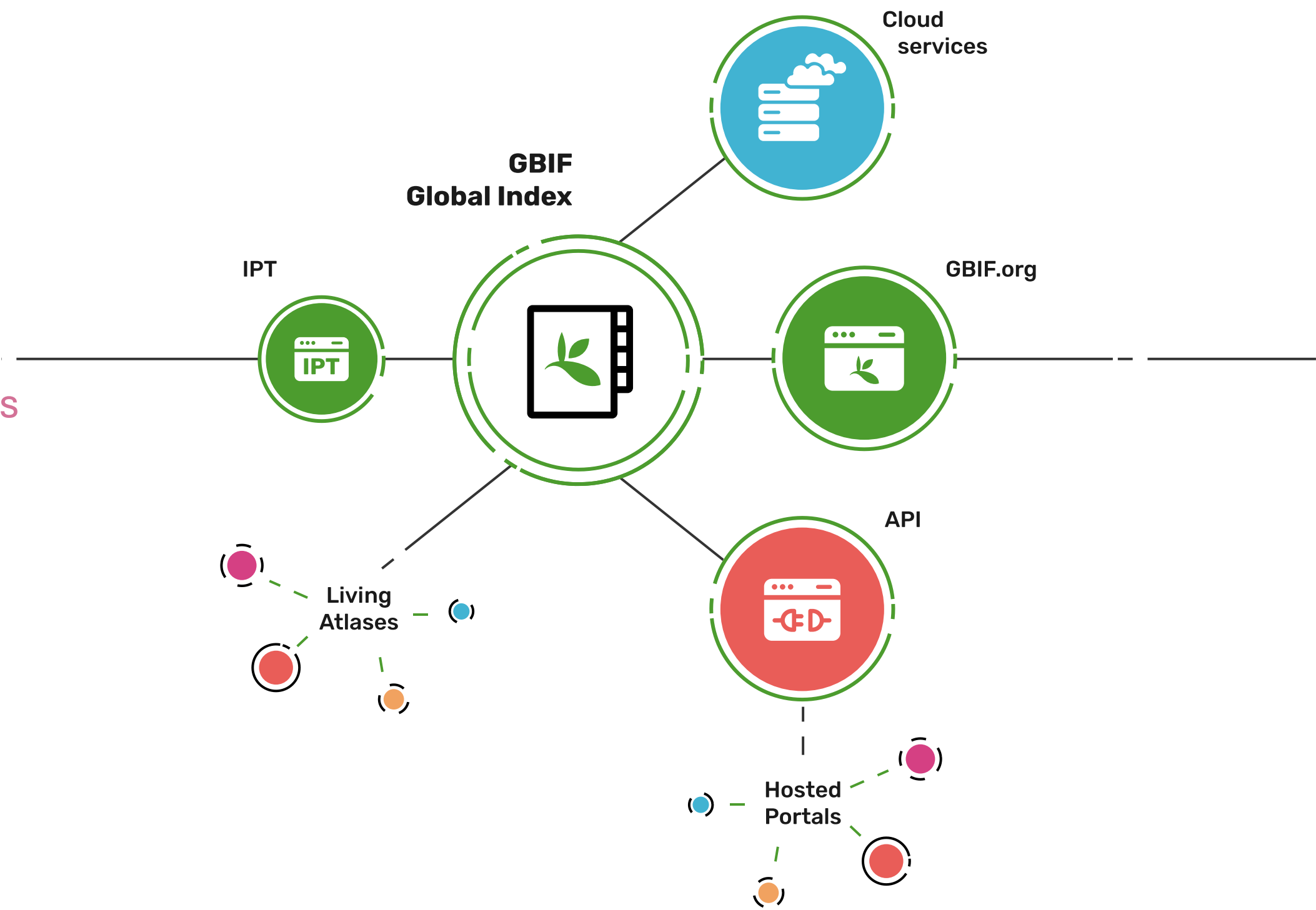
Publications from a search of the Web of Science within environmental domains papers using the words “environmental DNA” or “eDNA.” Publications have increased exponentially in recent years, demonstrating the growing popularity of this method. Colors represent Web of Science 'Research Areas', with 'Other' being the aggregation of everything that is not 'Environmental Sciences and Ecology.'



eDNA helps, but it's not a silver bullet



Share
FAIR and open access



eDNA at GBIF: how it is accessed and used



Get dataHow-toToolsCommunityAbout



Latest GBIF news on biodiversity and DNA



Invitation to share DNA metabarcoding data in data-publishing tool pilot



Updated guide adds details on how to publish DNA-derived data on marine life



Exploring eDNA data webinar recording and resources



Dig deep and share: Call for data papers describing datasets on soil biodiversity
EXTENDED

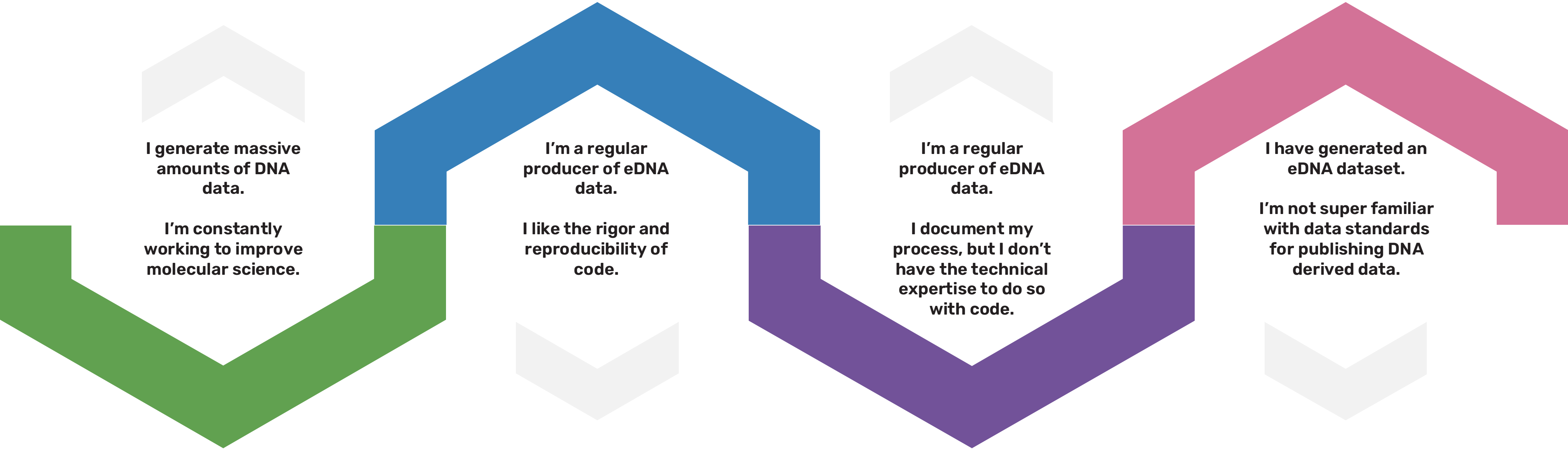




Spectrum of Use Cases



Spectrum of Approaches



eDNA at GBIF: how it is accessed and used



Global Biodiversity
Information Facility

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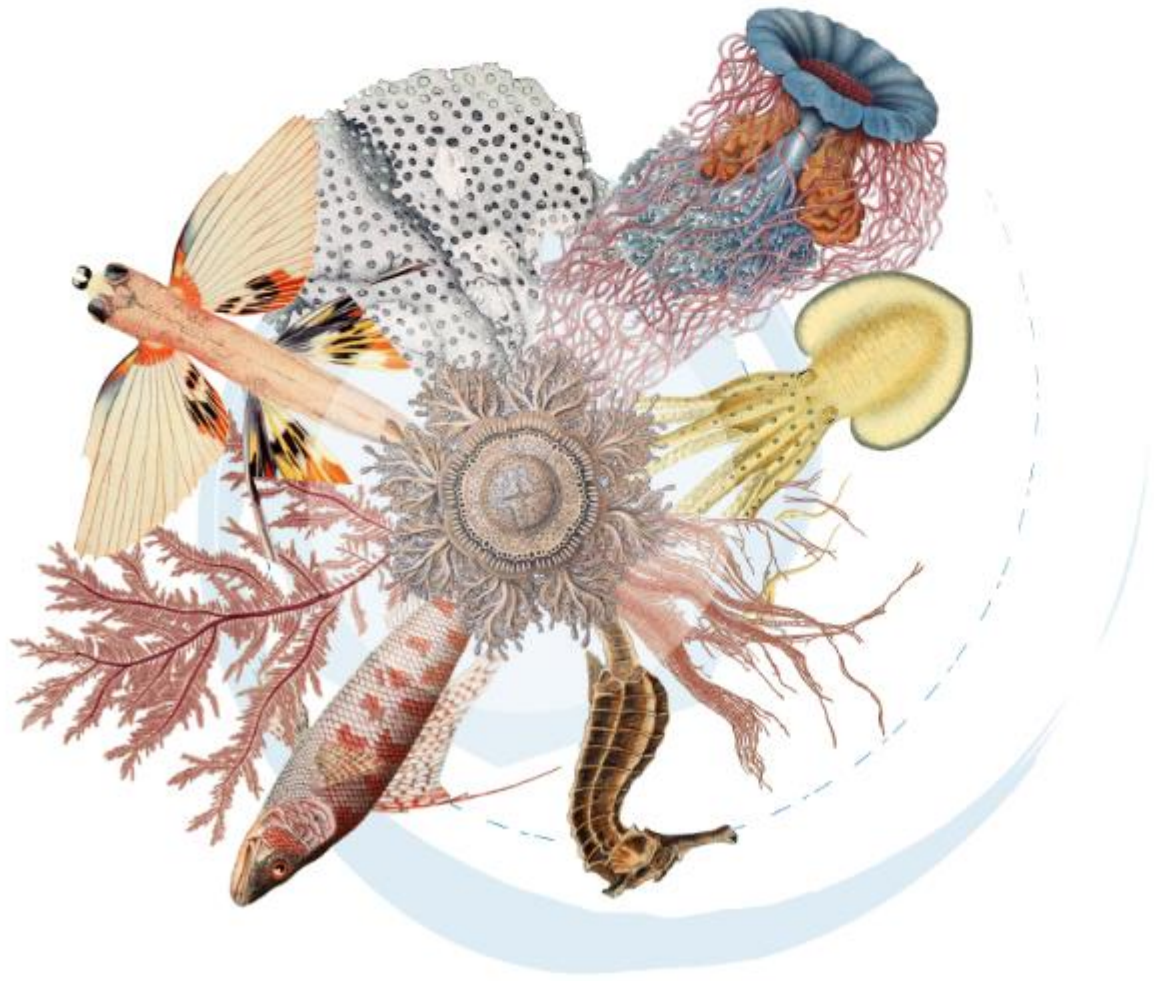
Preface

[1. Introduction](#)
[1.1. Rationale](#)
[1.2. Target audiences](#)
[1.3. Introduction to DNA-derived occurrence data](#)
[1.3.1. Environmental DNA as a source for DNA-derived](#)

Publishing DNA-derived data through biodiversity data platforms

Kessy Abarenkov · Anders F. Andersson · Andrew Bissett · Anders G. Finstad · Frode Fossey · Marie Grosjean · Michael Hope · Thomas S. Jeppesen · Urmas Kõljalg · Daniel Lundin · R. Henrik Nilsson · Maria Prager · Pieter Provoost · Dmitry Schigel · Saara Suominen · Cecilie Svenningsen · Tobias Guldberg Frøslev – Version 1.3.0, 7 June 2023

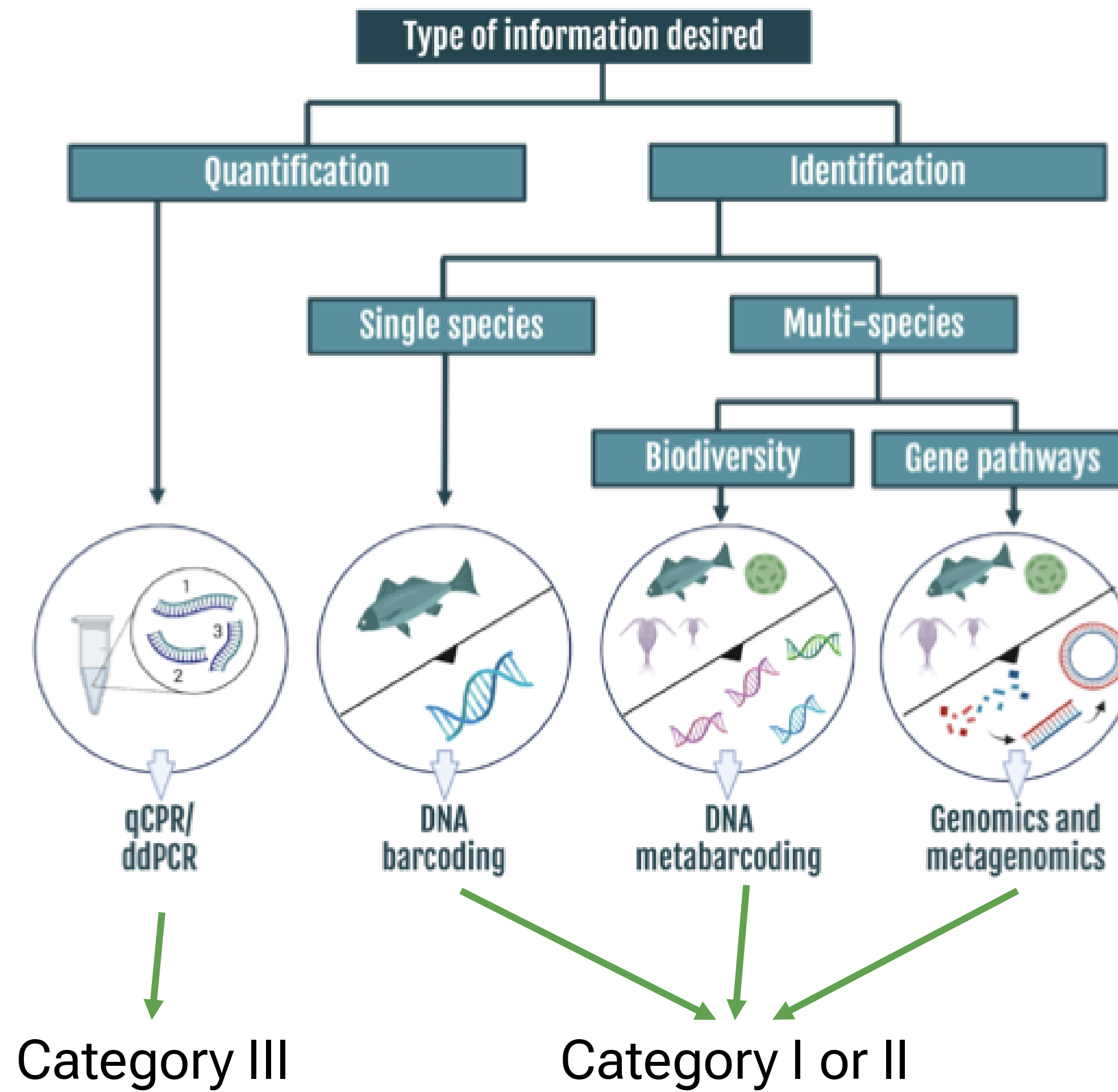
This document is also available in [PDF format](#) and in other languages: [español](#), [français](#), [简体中文](#).



<https://doi.org/10.35035/doc-vf1a-nr22>



Which eDNA approach is right for me?

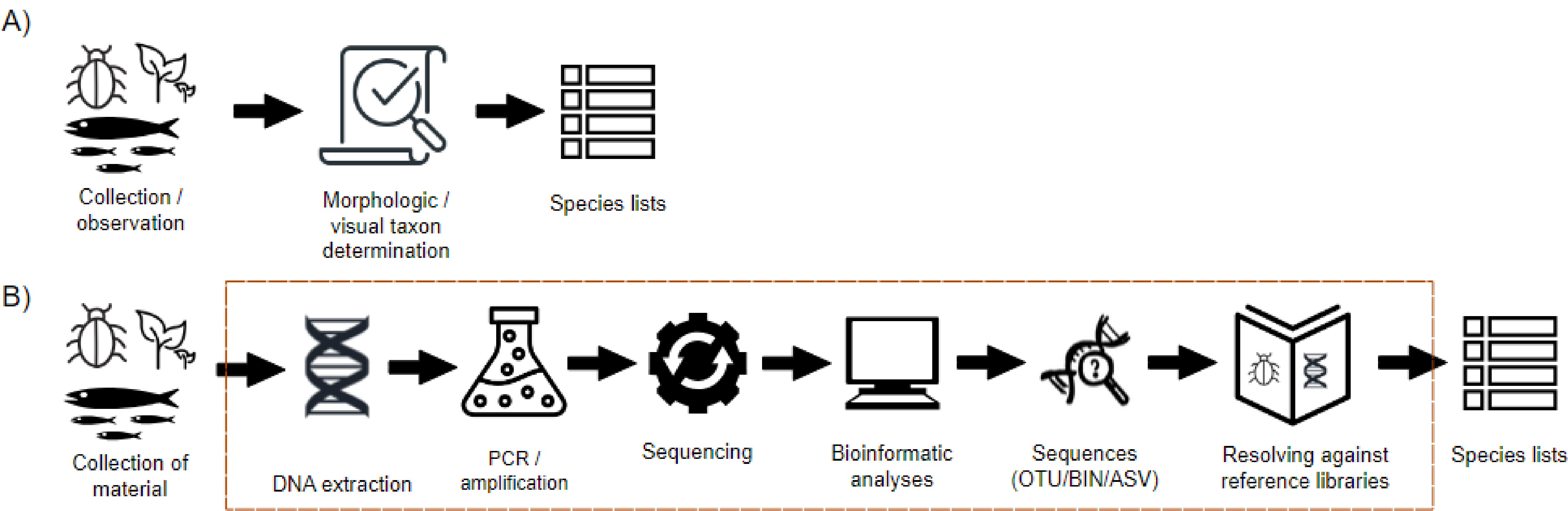


GBIF
DNA Publishing Guide
Category (Table 1):

<https://doi.org/10.35035/doc-vf1a-nr22>



The DNA-Derived Data Extension



Modified from Figure 2:

Abarenkov K, Andersson AF, Bissett A, Finstad AG, Fossøy F, Grosjean M, Hope M, Jeppesen TS, Kõljalg U, Lundin D, Nilsson RN, Prager M, Provoost P, Schigel D, Suominen S, Svenningsen C & Frøslev TG (2023) Publishing DNA-derived data through biodiversity data platforms, v1.3. Copenhagen: GBIF Secretariat. <https://doi.org/10.35035/doc-vf1a-nr22>.



GBIF eDNA data converter tool

 eDNA Tool - Test environment

[New dataset](#) [FAQ](#) [Login](#)

eDNA data converter

bridging metabarcoding and biodiversity

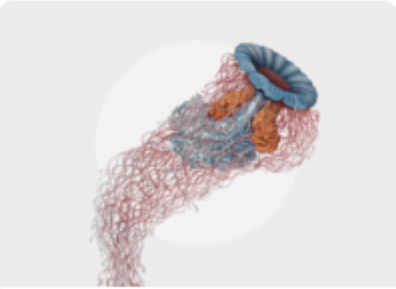




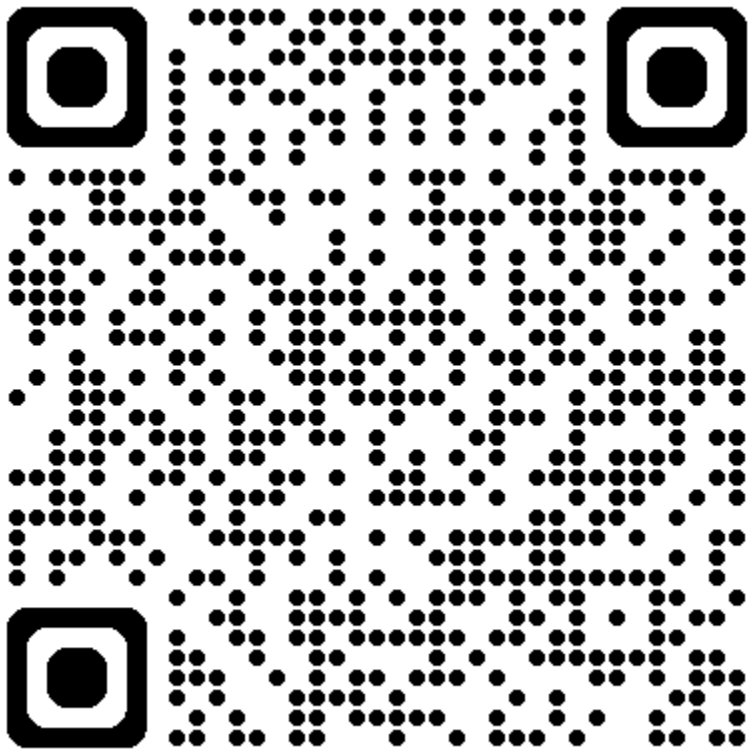
How to use it?



Got feedback?

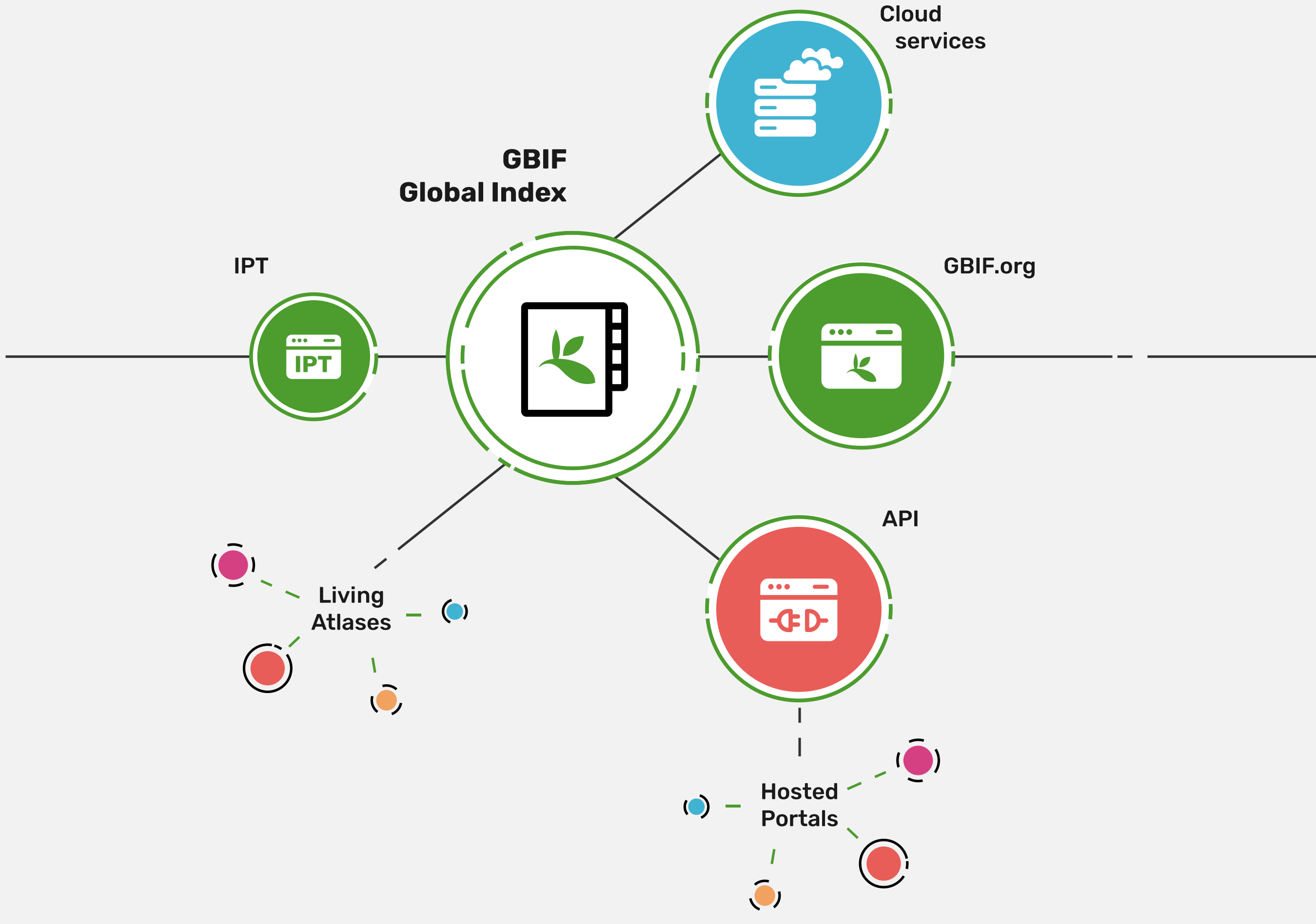


FAQ



Access to biodiversity evidence

Share
FAIR and open access



Example published datasets that use the DNA-derived data extension



Museum Specimen + DNA extension

Example Occurrence:

<https://www.gbif.org/occurrence/4445072107>

Dataset Link:

<https://www.gbif.org/dataset/040c5662-da76-4782-a48e-cdea1892d14c>



Exploring marine biodiversity with 16S + 18S

Example Occurrence:

<https://www.gbif.org/occurrence/444507210>

Dataset Link:

<https://www.gbif.org/dataset/9012def0-bd87-48a0-ac9e-e0e78dd37689>



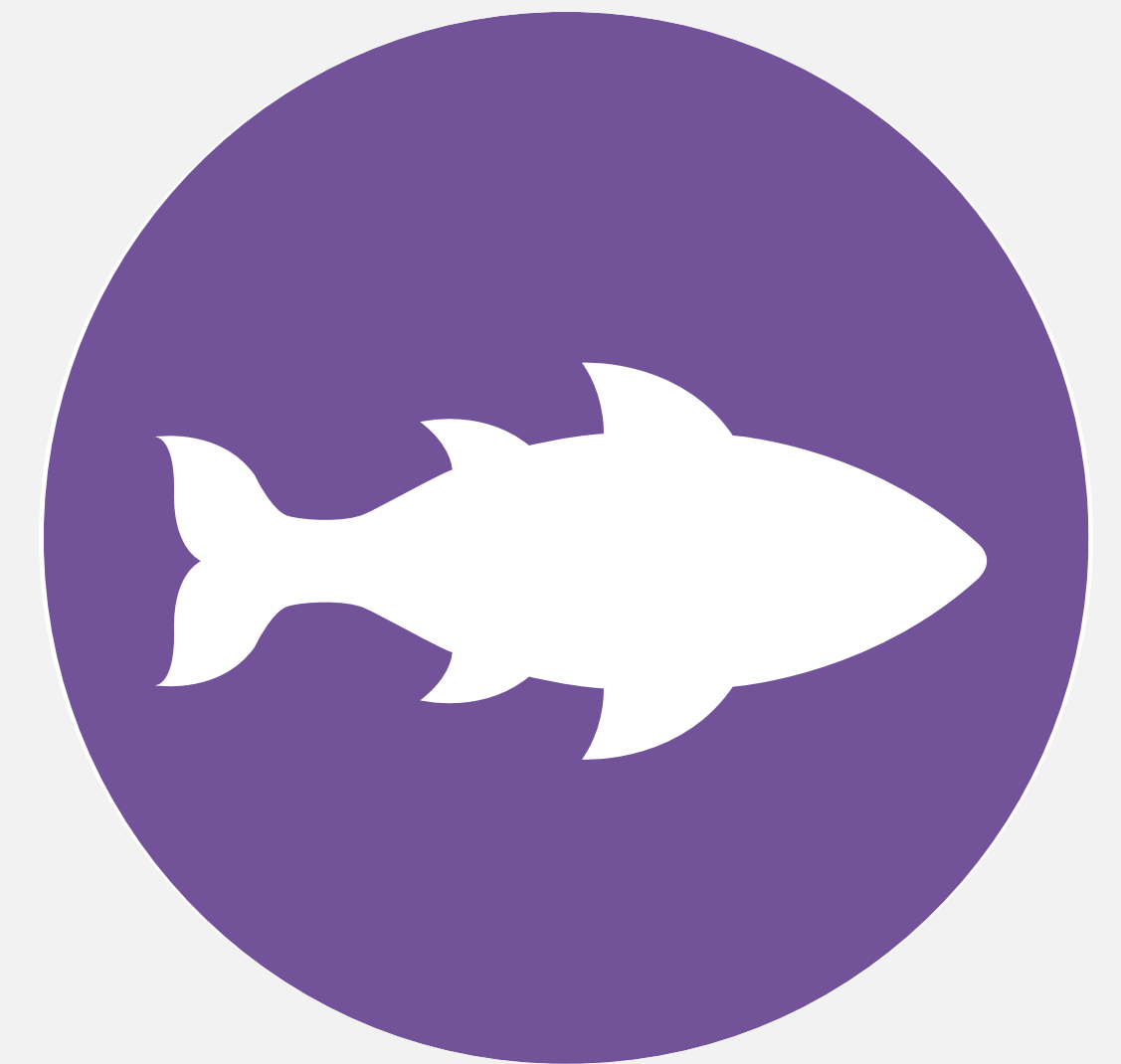
COI eDNA of intertidal zone

Example Occurrence:

<https://www.gbif.org/occurrence/4537370216>

Dataset Link:

<https://www.gbif.org/dataset/9358fbd7-cfd0-4eab-99fa-0934396a0529>



qPCR + metabarcoding of invasive species

Example Occurrence:

<https://www.gbif.org/occurrence/4406079323>

Dataset Link:

<https://www.gbif.org/dataset/b88e40a8-c39e-4f8f-962a-7f6d93a977a4>

